UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

STAT 576: Empirical Process Theory and Weak Convergence — Fall 2025
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Lecture 16

Content. Content/keywords of lecture (very briefly)

1. Proof of Efron-Stein Inequality
2. Uniform Law of Large Numbers/Empirical Risk Minimization

1 Proof of Efron Stein Inequality

Theorem 1. *(Efron-Stein Inequality)* Let $X_1, \dots, X_n$ be independent RVs, $f : \mathbb{R}^n \to \mathbb{R}$ is a function. Then for $X = (X_1, \dots, X_n)$,

$$\mathrm{var}(f(X)) \leq \frac{1}{2} \sum_{i=1}^{n} \mathbb{E}[(f(X) - f(X^{(i)}))^2]$$

where $f(X^{(i)}) = f(X_1, \dots, X_{i-1}, X_i', X_{i+1}, \dots, X_n)$ and X_i' is an independent copy of X_i.

Proof: Assume wlog $\mathbb{E}[f(X)] = 0$. Define the Doob martingale

$$M_i = \mathbb{E}[f(X)|X_1, \dots, X_i]$$

$$M_0 = \mathbb{E}[f(X)] = 0 \quad M_n = f(X).$$

Write $f(X) = \sum_{i=1}^{n} D_i$ where $D_i = M_i - M_{i-1}$. Then,

$$\mathrm{var}(f(X)) = \mathbb{E}[f(X)^2] = \mathbb{E}[(\sum_{i=1}^{n} D_i)^2] = \mathbb{E}[\sum_{i=1}^{n} D_i^2] + 2\mathbb{E}[\sum_{1 \leq i < j \leq n} D_i D_j].$$

Claim 1. $\mathbb{E}[\sum_{i<j} D_i D_j] = 0$

Proof of Claim: $\mathbb{E}[D_i D_j] = \mathbb{E}[\mathbb{E}[D_i D_j|\mathcal{F}_i]]$, where $\mathcal{F}_i = \sigma(X_1, \ldots, X_i)$ by the tower property. Since D_i is measurable with respect to $\mathcal{F}_i$,

$$\mathbb{E}[\mathbb{E}[D_i D_j|\mathcal{F}_i]] = \mathbb{E}[D_i \mathbb{E}[D_j|\mathcal{F}_i]] = 0$$

since $\mathbb{E}[M_j|\mathcal{F}_i] = M_i \quad \forall j \geq i$.
We bound the sum $\mathbb{E}[\sum_{i=1}^n D_i^2]$.

$$\mathrm{var}(f(X)) = \sum_{i=1}^n \mathbb{E}[(M_i - M_{i-1})^2] = \sum_{i=1}^n \mathbb{E}[(\mathbb{E}[f(X)|X_1, \ldots, X_i] - \mathbb{E}[f(X)|X_1, \ldots, X_{i-1}])^2]$$

$$= \sum_{i=1}^n \mathbb{E}[(\mathbb{E}_{X_{i+1:X_n}}[f(X)|X_1, \ldots, X_i] - \mathbb{E}_{X_{i+1:X_n}}[\mathbb{E}_{X_i}[f(X)]|X_1, \ldots, X_i])^2]$$

$$= \sum_{i=1}^n \mathbb{E}[(\mathbb{E}_{X_{i+1}:X_n}[f(X) - \mathbb{E}_{X_i}[f(X)]|X_1, \ldots, X_i])^2]$$

By Jensen's inequality

$$\mathrm{var}(f(X)) \leq \sum_{i=1}^n \mathbb{E}_{X_1:X_n}[\mathbb{E}_{X_{i+1}:X_n}[(f(X) - \mathbb{E}_{X_i}[f(X)])^2|X_1, \ldots, X_i]]$$

$$= \sum_{i=1}^n \mathbb{E}_{X_1:X_n}[(f(X) - \mathbb{E}_{X_i}[f(X)])^2] = \sum_{i=1}^n \mathbb{E}_{X_1:X_{i-1}, X_{i+1}:X_n}[\mathbb{E}_{X_i}[(f(X) - \mathbb{E}_{X_i}[f(x)])^2]]$$

$\mathbb{E}_{X_i}[(f(X) - \mathbb{E}_{X_i}[f(x)])^2]$ is the variance of $f(X)$ along X_i. Using the fact that $\mathrm{var}(z) \leq \frac{1}{2}\mathbb{E}[(z - z')^2]$, we have

$$\mathrm{var}(f(X)) \leq \sum_{i=1}^n \mathbb{E}_{X_1:X_{i-1}, X_{i+1}:X_n}[\mathbb{E}_{X_i}[(f(X) - \mathbb{E}_{X_i}[f(x)])^2]] \leq \frac{1}{2}\sum_{i=1}^n \mathbb{E}_{X_1:X_n}[(f(X) - f(X^{(i)}))^2]. \quad \square$$

Remark 1. Since X_i's are independent, the joint law of $X_1, \ldots, X_n$ is a product measure, so we can integrate in whichever order we like.

2 Towards ULLN

- A key step in statistic

  Controlling the supremum of a collection of RVs, either in expectation or with high probability

- $\sup_n |z_n|$, z_n's are sub-Gaussian

GOAL: Do this for "arbitrary"/infinite classes.

$$\sup \mathbb{E}[|z_n| = \frac{2}{\pi}] \quad z_n \sim \text{iid } \mathcal{N}(0,1)$$

$$\mathbb{E}[\sup_n |z_n|] = \Theta(\sqrt{\log N}) \quad n = 1, \ldots, N.$$

In general, by Fatou's lemma and Jensen inequality

$$\sup_n \mathbb{E}[|z_n|] \leq \mathbb{E}[\sup_n |z_n|].$$

'$\Theta(\sqrt{\log N})$' is the "price" of putting sup inside the expectation.

Uniform: "Supremum" of a collection inside expectation.

3 Kolmogorov's Goodness-of-Fit Test

Suppose we have $X_1, \ldots, X_n \sim \text{iid } F$. We'd like to hypothesis test between $H_0 : F = F_0$ and $H_1 : F \neq F_0$.

$$F_n(\theta) := \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}\{X_i \leq \theta\} \quad F_n(\theta) \overset{\text{a.s.}}{\to} F(\theta) \quad \text{[SLLN/WLLN]}$$

Definition 1. *(Kolmogorov-Smirnov Statistics)*

$$D_n := |F_n(\theta) - F(\theta)|$$

Theorem 2. *(Glivenko-Cantelli)*

$$D_n \overset{\text{a.s.}}{\to} 0.$$

This theorem provides a uniform guarantee as we sup over all θ.
$\sup_\theta |F_n(\theta) - F_0(\theta)|$ holds under H_0 but not under H_1. We need to study the asymptotic distribution of D_n.

Theorem 3. *(Kolmogorov '33)* [Assume f is continuous]

1. The asymptotic distribution of D_n does not depend on the distribution of X_i [Universality].

   Idea:

   $$\sup_\theta |\frac{1}{n}\sum_{i=1}^n \mathbb{1}\{X_i \le \theta\} - F(\theta)| \stackrel{d}{=} \sup_\theta |\frac{1}{n}\sum_{i=1}^n \mathbb{1}\{F(X_i) \le F(\theta)\} - F(\theta)|$$
   $$= \sup_{t\in[0,1]} |\frac{1}{n}\sum_{i=1}^n \mathbb{1}\{U_i \le t\} - t|$$

   where $U_1, \dots, U_n \sim$ Unif$[0,1]$.

2. D_n is of order $\frac{1}{\sqrt{n}}$ $\quad (D_n = O(\frac{1}{\sqrt{n}}))$.

   Idea: DKW inequality: $\mathbb{P}[D_n \ge t] \lesssim \exp(-ct^2 n)$. $t = \Theta(\frac{1}{\sqrt{n}})$ is the right order.

3. $\sqrt{n}D_n \to \sup_{t\in[0,1]} |B(t)|$, Brownian bridge.
   ($\mathbb{P}[\sqrt{n}D_n \le 2.4] \approx 0.999973$) asymptotically.

Our Focus: Non-asymptotic, finite sample guarantee.
Sometimes "sup" appears in disguise:

$$\mathbb{E}[||z||] = \mathbb{E} \sup_{u:||u||_2=1} \langle u, z\rangle. \quad \text{(variational characterization of norm)}$$

4 Elements of Statistical Learning Theory

Suppose that $S = \{(X_1, Y_1), \dots, (X_n, Y_n)\}$ is a dataset (training data).

Assumption 1.

- $(X_i, Y_i) \sim$ *iid* $P_X \times P_{Y|X}$
- $X_i \in \mathcal{X}, \quad Y_i \in \mathcal{Y}$
- $f \in F = \{f : \mathcal{X} \to \mathcal{Y}\}$ *($\mathcal{F}$ can be the class of decision trees, neural networks, linear classifiers, etc.)*
- *Loss Function* $\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}_{\ge 0}$

5 Empirical Risk Minimization

Definition 2. *(Empirical Risk)* For $f \in \mathcal{F}$, define

$$\hat{L}(f) = \frac{1}{n}\sum_{i=1}^{n} \ell(f(X_i), Y_i)$$

ERM: $\inf_{f\in\mathcal{F}} \hat{L}(f)$

Example 1.

- *Linear Regression:*

$$\mathcal{F} = \{x \to \langle w, x\rangle : w \in \mathbb{R}^d\} \quad \ell(a,b) = (a-b)^2$$

- *Classification:*

$$\mathcal{F} = \{x \to \frac{sgn(\langle w, x\rangle) + 1}{2} : ||w||_2 = 1\} \quad \ell(a,b) = \mathbb{1}\{a \neq b\}$$

Population Loss/ Generalization Error:

$$L(f) = \mathbb{E}_{(X,Y)}[\ell(f(X), Y)]$$

here (X, Y) is the fresh sample drawn indepently from the distribution generating S.